Draft Rules for Evaluating “Practical Trailing” Competitions

Philosophy of the Rules: Toward Operational Practice

These rules aim to establish a methodology that best evaluates the **operational efficiency** of the team. The objective is not merely to create a sport, but to develop the most precise tool for verifying a team's ability to locate a subject in real-world environments.
Instead of measuring absolute time, insisting on precise tracking of the trail layer's footsteps, or locating dropped articles, we assess the **team's efficiency** and the **handler's ability to read their dog**. The aim is not sporting competition, but a verification of skills under stress levels that simulate real-world operations.
A core principle of these rules is that the optimal strategy for winning the competition is identical to the best preparation for a search and rescue mission. Points are awarded for **measurable performance metrics** that save lives in the field: rapid subject location, intensity of work, and the accurate interpretation of the dog’s behavior.
Why hold competitions when certifications exist? A standard certification provides a binary outcome—a team either passes or fails. While this is valuable information, it is diagnostically limited: it does not reveal exactly what needs improvement. This competition’s scoring system provides **detailed feedback**, allowing the handler to identify precisely in which phase of the trail and in which specific criteria the team fell short. As such, these competitions are not an alternative to certifications but a supplement: while certification verifies basic proficiency, the competition highlights the path toward further development.
Objective Assessment: Eliminating judge subjectivity is complex; therefore, the evaluation is conducted via a dedicated application that analyzes GPS logs from both the trail layer and the K9 team.

Primary Evaluation Criteria[1]

1. Finding the subject
2. Work efficiency
3. Trailing accuracy
4. Handler’s ability to read their dog

Organisation of Performance

The team starts with a scent article. The dog’s task is to locate and follow the scent trail of the subject and lead the handler to its end — to the subject.

There may be several items belonging to the subject along the trail. Finding an item confirms the correctness of the trail. Items should be inconspicuous, easily mistaken for “litter” that might be found in that area (a paper tissue, a beer can, etc.).

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[1] *Detailed justification of criteria: see section "Expert Justification of Evaluation Criteria."*

Competitions must be double-blind — not even the judge (!!) may know the trail layer's route. The judge monitors the handler's work, records reported checkpoints (see below), and ends the run when the time limit is reached. After the limit expires, the handler may still report one final checkpoint.

Trail Parameters

For fairness, all trails should be approximately:

- the same length and age,
- run through similar terrain,
- with the same number of sharp turns, loops, blind alleys, scent pools, etc.,
- with the same number of items, placed at equal intervals along the trail.

All trails should ideally be laid downwind or crosswind. Any items serve only as confirmation points for the handler (a checkpoint may be called at them).

Example Scoring

Point values for each section may be adjusted by the competition organiser to emphasise one or more aspects of evaluation. This is an example only!

1. Finding the subject

0 or 100 points

2. Work efficiency

+20 points per km/h (100 points at 5 km/h). Speed is calculated from the trail layer's route length between the start and the last acknowledged checkpoint (or subject find), divided by the elapsed time from team start. The result is multiplied by a weight based on the successfully completed portion of the trail — e.g. if the last checkpoint is at the halfway point, weight = 0.5, giving 10 points per km/h.

3. Trailing accuracy

Maximum 100 points. Evaluated using weight w based on the dog's deviation from the trail layer's route. The weight w changes continuously according to a mathematical function which at key points takes the following values:

Deviation d	Weight w	Verbal assessment
< 5 m	1.0	Excellent accuracy
15 m	0.5	Acceptable accuracy
30 m and above	0.1 and below	Inaccurate

For each segment of the dog's track, the distance to the nearest point on the trail layer's route is calculated, and a weight function *w* is applied to this distance. The resulting accuracy score is a weighted average of these weights – where each segment is weighted both by its length and by the time spent at that location. The average of these two values forms the final score. In this way, large deviations are penalized more heavily than small ones – a dog that consistently stayed 15 m from the

trail scores 50 points, but a dog that followed the trail perfectly for half the route and then deviated 30 m for the other half scores less than 50 points, because the weight function is non-linear.

4. Handler's ability to read their dog — checkpoints

What is a checkpoint: The handler may at any point during the run call a "checkpoint" — a moment where they are confident the dog is actively working the scent trail. The judge acknowledges the checkpoint only if the subject is not yet in sight.

How to call one: Whenever you are confident the dog is working correctly, say to the judge: **"Checkpoint."** That is all — the scoring maths is handled automatically by the application.

Acknowledgement rules:

- A checkpoint may be called at any time during the trail run, repeatedly.
- The judge acknowledges the checkpoint only if the subject is not yet in sight.
- A checkpoint is also recorded at an item reported by the handler (regardless of whether the dog indicated it — the handler's call is decisive).
- Finding the subject does not count as a checkpoint — it is scored separately under criterion 1.
- The end of the run (where the handler stopped or the time limit expired) counts as a checkpoint only if the handler explicitly calls it and the subject is not yet in sight.

Using checkpoints to return to the trail: If the dog loses the trail, the handler may ask the judge to return to the last called checkpoint. The handler's flanker may fulfil this role if the handler has one.

How checkpoints are scored:

The scoring rewards handlers who read their dog consistently throughout the entire trail, over those who call only one or two checkpoints near the end. At the same time, adding a further checkpoint can never reduce the score achieved (provided the point is near the trail layer's route).

Each checkpoint contributes to the total score based on two factors:

- **Point accuracy —** how close to the trail layer's route the team was at the moment of calling (same weight function as in criterion 3).
- **Interval length from the previous checkpoint —** a long gap (the handler went a long time without calling — was uncertain whether the dog was on trail) is penalised. Approximately: if the distance between handler-called checkpoints is 50 % of the trail length, the team receives only 50 % of the available points in this category.

Maximum points in this category: 100.

Expert Justification of Evaluation Criteria

Criterion 2: Work efficiency

Work efficiency is a significant evaluation criterion, but it does not measure pace as a sporting performance. It measures something else: how quickly the team delivered quality work on the completed portion of the trail. The literature is unambiguous that the speed of locating a missing person has a direct impact on survival.

Time as a life-saving factor

Luboš Satora states the fundamental principle of mantrailing in the context of real deployments unequivocally:

> *“After all, what everyone hopes for in the end is just one thing: finding the (living) missing person as quickly as possible.”*
>
> *(Satora, Základy mantrailingu)*

Satora supplements this by adding a practical deployment rule: “In a deployment, the rule always applies: ‘The sooner, the better.’” Speed of finding is not a sporting ambition — it is a direct consequence of how efficiently the team works on the trail.

Csaba Granicz documents a concrete case where the time factor was critical — a missing man who had swallowed large amounts of medication before leaving home:

> *“He had taken a lot of medication before getting in the car, so the time factor was critical.”*
>
> *(Granicz, Mantrailing Fine Tuned)*

Satora further confirms the urgency in his description of the emergency response system:

> *“Late notification of the team significantly reduces the success of their search.”*
>
> *(Satora, Základy mantrailingu)*

While this refers to delay in calling the team rather than the team’s own working pace, the logic is the same: every minute matters. In cases involving medication overdose, hypothermia, or suicidal intent — all situations documented by Satora and Granicz — time is literally life-or-death.

Direct response to Satora’s criticism

Satora writes explicitly: “Mantrailing is not a sport, there is no need to compare ourselves with each other. The criterion is just one — finding the missing person as quickly as possible, without any additional conditions.” This objection is legitimate. If we dismiss it, the rest of the rules lose their footing. It must be addressed directly.

Satora’s criticism targets competitions where speed is the sole or dominant criterion — where the team that arrived first wins, regardless of how the dog worked. In such a format, Satora is absolutely right: we are rewarding sprinting, not trailing. These rules, however, do not measure speed as a sporting performance. Speed here is conditional on the dog’s work: it is calculated exclusively from the length of trail the team covered with acknowledged checkpoints. A team without acknowledged checkpoints — a team where the handler cannot demonstrate that the dog was working throughout — gains nothing for speed. Speed here is the result of quality work, not its substitute.

Speed as part of a composite evaluation

Work efficiency is evaluated as part of a broader system — not as the sole or dominant criterion. A team that moves fast but works inaccurately and fails to read the dog will overall lose to a team that moves slower but delivers quality work. This is a fundamental difference from competitions where time is the only criterion. Speed is rewarded here only when it is supported by a completed trail section and acknowledged checkpoints — that is, by actual work.

Summary: what the criterion measures

- **Reflects** the core principle of mantrailing as stated by Satora: the goal is to find the living missing person as quickly as possible. Scoring efficiency motivates teams to work effectively — in a real deployment, every saved minute can mean the difference between life and death.

- **Measures** speed as the result of quality work — the calculation is based on the length of trail completed and acknowledged checkpoints, not on raw pace. A fast team without acknowledged checkpoints or without a find gains no advantage.
- **Does not disproportionately reward** speed at the expense of quality — the maximum weight of this criterion is comparable to the other criteria, and a team that moves fast but works inaccurately will not benefit in the overall score.

Criterion 3: Trailing accuracy

Evaluating trailing accuracy is one of the most sensitive points of the rules, as opinions in the mantrailing community differ on whether accuracy is a measurable and relevant indicator at all. This section summarises the expert arguments underpinning the criterion.

Mantrailing accuracy is not the same as sport-tracking accuracy

The most common objection to evaluating accuracy is: "a mantrailing dog simply does not work precisely along the trail layer's route." This objection is correct — but it targets a different kind of evaluation than these rules use. Luboš Satora explains the physics of scent particles:

> *"Under ideal conditions, these scent particles could indeed land precisely where the missing person walked. In most cases, however, they are carried by the wind to places where they settle on natural obstacles. This explains the sometimes seemingly incomprehensible movement of the dog, which may work several tens of metres from the actual route the person took."*
>
> *(Satora, Základy mantrailingu)*

Csaba Granicz states this distinction even more explicitly and defines what accuracy in mantrailing actually means:

> *"When training your mantrailing dog, it is important to keep the tracking as accurate as possible — that is, to keep the so-called wavering to a minimum and to keep the dog as close as possible to the scent residues left behind. Here, however, accuracy is not measured in terms of the path the trail layer has physically travelled, but rather by trying to estimate, to the best of our ability, the external factors that influence the spread of odors. So, we do not expect our dog to follow the physical trail to the nearest centimeter. We must leave him much more room to work the scent. However, this freer style of tracking carries with it the risk that if the dog works way too wide and does not try to get close to the scent residue of the trail, his work will become inaccurate, and he may easily lose the trail."*
>
> *(Granicz, Mantrailing Fine Tuned)*

Accuracy in mantrailing is therefore not replication of the trail layer's physical route (that is sport tracking), but the degree to which the dog remains within reach of the scent trail and does not unnecessarily drift into areas where scent is absent. This is why these rules treat deviations under 10 m as "excellent accuracy" and only begin to apply significant weight reduction from 50 m upwards.

Large deviations as an operational risk: empirical evidence

Granicz's "freer style" is not without risk — and Granicz himself flags this (see quote above). Jeff Schettler documents this principle through a concrete field case: his dog began working too far from the trail — due to a strong scent cone drifting from a building — lost the connection, and the deployment ended in failure:

> *“If the spot drifted far away from the trail, my dog would eventually forget where the trail was in favor of all the fresh odor that led nowhere. […] What happened in the real case I wrote about was we struggled so long and hard, and I corrected my dog so much that he shut down and the trail did not exist anymore for those reasons alone. I should have been spit on; I was so annoyed with myself!”*
>
> *(Schettler, The Tao of Trailing)*

Schettler thus empirically demonstrates that a dog working too far from the trail can lose contact with the scent entirely — even when the scent physically exists. The deployment then depends on the handler’s memory, not the dog.

Informational value of accurate work: the last known scent point

A dog working accurately that reaches the point where it can no longer follow the trail provides the rescue service with a key piece of information: where the target scent was last present. This is critical for follow-on teams or other search resources. Granicz notes the importance of accurate GPS records directly in a deployment context:

> *“GPS tracking is also a useful aid, giving you instant warning if you plan to lead your dog back to the wrong place. […] On more than one occasion during our training sessions, a dog handler has noticed that his dog is no longer following the trail of the person hiding, so he leads him back to where he thinks they were. But that point was up to ten meters further away from the actual trail. In such cases, the dog may be lucky enough to ‘air scent’ the trail […] but it may not be able to do so and the search may end in a dead end because the dog handler’s mind map was incorrect.”*
>
> *(Granicz, Mantrailing Fine Tuned)*

Summary: what the criterion measures and does not measure

- **Does not penalise** the dog for failing to work centimetre-precisely along the trail layer’s physical route. Deviations under 10 m are rated as excellent — consistent with normal scent dynamics.
- **Penalises** large deviations (50 m and above), which according to the literature (Granicz, Schettler) signal loss of scent contact and significantly increase the risk of deployment failure.

Criterion 4: Handler’s ability to read their dog — checkpoints

Reading the dog — the handler’s ability to correctly interpret the dog’s behaviour and at every moment distinguish whether the dog is working on the scent trail or not — is unanimously regarded in the literature as the key competency of the entire mantrailing team. Checkpoints in this competition are a measurable tool that objectively captures this competency.

Reading the dog as a core handler skill

Luboš Satora defines reading the dog as the handler’s primary task throughout the entire trail:

> *“The handler’s main task when working the dog is to correctly and objectively analyse its movements and behaviour and translate them into human language. Whoever learns this and gains experience from long-term observation of the dog’s movements has the key to success.”*
>
> *(Satora, Základy mantrailingu)*

Satora further lists the inability to read the dog as one of the most common handler errors: “They cannot read from the dog’s movements and behaviour what it is doing.” This inability is not merely a personal weakness — it is a systemic barrier to the entire team’s effective performance.

Schettler states this requirement even more radically. He describes how a handler lacking this competency inevitably takes control — and in doing so, damages the dog:

> *“Many handlers make the mistake of allowing a dog that is not actively following scent to get to the area where the subject of their search is hiding — to allow them to make the find. They do this because they cannot read the dog’s body language that screams: ‘NOT FOLLOWING SCENT!’ […] The dog will learn a behaviour that the handler unintentionally teaches it — if the dog learns that when the chips are down the handler will lead it to the subject’s location, it will learn to ‘not work’ each and every time, and simply run.”*
>
> *(Schettler, The Tao of Trailing)*

In other words: a handler who cannot read the dog gradually destroys the dog’s ability to work independently. In a competition context this effect will not show immediately — but that is precisely why it must be measured.

Checkpoints as an operational tool — not merely a competition element

Granicz describes how this ongoing mental map of points where the handler was confident in the dog’s work is used directly in real deployments:

> *“During our deployments, it’s vital to constantly update and evaluate the mental map we keep in our heads. Exactly what route we’ve taken so far, where we’ve experienced unusual conditions. Where our dog has shown behavioural changes, how far he has confidently followed the trail and where is the point where we are no longer confident in our four-legged companion’s work. […] The mind-map allows you to make adjustments on the go and provides useful information for the head of the operation during the post-evaluations of the tracking.”*
>
> *(Granicz, Mantrailing Fine Tuned)*

Granicz further describes a case where it was precisely the logging of points where the dog changed behaviour that led to finding a missing woman alive in a grove — even after the dog had previously indicated the end of the trail. He also explains why this information is valuable to the operations commander:

> *“Also typical of these two applications is the possibility for the dog handler to ask the flanker to place so-called markers on the map. These points are placed where the dog is moving along the track without stopping, but the handler notices a change in the dog’s behaviour. Sometimes a dog may not stop at a critical point for some reason, but its reaction may suggest that if the tracking fails, it is worth returning to that point and scanning the surrounding area with a mantrailer or an air scent dog. We have already found missing persons in this way.”*
>
> *(Granicz, Mantrailing Fine Tuned)*

Kocher emphasises that it is precisely the ability to read the dog that constitutes the “10 %” the handler is responsible for, and which determines whether a correctly working dog brings the deployment to a successful conclusion:

> *“A dog team’s success is determined: 90 % by the dog’s ability, 10 % by the handler’s ability. Because of that 10 %, you need to practice helping your dog. […] One common weakness displayed by dog handlers is the inability to react when their dog loses the*

> *trail. […] By reading the negative indications you will have a general idea where the dog last had the scent."*
>
> *(Kocher, How to Train a Police Bloodhound)*

Ongoing calling as a quality measure, not merely a count of reported points

The scoring system rewards handlers who read their dog consistently throughout the entire trail — not only those who call one or two checkpoints just before the end. This approach derives directly from how Granicz describes fieldwork:

> *"During our training, we cannot stress enough the importance of keeping a mind-map during each tracking session — i.e., making a mental note of where you have been on the track and where you have noticed even slightly unusual behaviour in your dog as the success of a real search can easily depend on it."*
>
> *(Granicz, Mantrailing Fine Tuned)*

Competition checkpoints are therefore the translation of this mental map into measurable form: a handler who consistently and reliably calls the points where they are confident in the dog's work demonstrates precisely the competency that is most valuable in a real deployment.

Summary: what the criterion measures

- **Measures** the handler's ability to consistently and reliably identify when the dog is working on trail — a competency that the entire expert consensus (Satora, Schettler, Granicz, Kocher) regards as the handler's primary task.
- **Rewards** continuous dog-reading throughout the trail, not merely a chance match at the end — exactly as Granicz describes a functional mental map in a live deployment.
- **Provides** the rescue system with exceptionally valuable information: where the missing person was, which locations they passed through with certainty — information that can be passed on to a follow-on team or the operations commander (Granicz, Kocher).